

Collaboration and Generative AI Disclosure

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I worked on this assignment independently. However, I did use generative AI, specifically the Claude Opus 5 model, as a learning and support tool.

Since this was my first time working with Android Studio, I used the AI model extensively to help me understand the basic project structure after creating an Empty Activity project. I asked it to explain best practices for setting up different components in an Android application, how common Android templates are structured, and which UI components would be suitable for the design of my application.

I also used the model to help with debugging whenever I encountered errors. Although it generated some code suggestions, I did not simply copy everything it produced. I reviewed the suggestions and only used the parts that I felt would improve the application. I also made sure to understand all of the code suggested by the AI tool and how each part was being used within my application.

For example, the model suggested placing the Previous and Next navigation buttons below the OutlinedTextField, submit button, and information section because navigation controls are often placed near the bottom of a mobile interface. However, I decided to keep the Previous and Next buttons near the dimension details instead. I felt this made more sense because those buttons are specifically used to move between the different mobility dimensions and are not related to the OutlinedTextField section.

I verified the code by building and running the application after making changes, which helped me catch errors early. I also used Compose Preview to check the layout and UI before running the full app. I manually tested the Previous and Next buttons, input validation, and error messages to make sure everything worked correctly. I also reviewed Android Studio warnings and made sure I understood the code before keeping it in the final project.

Prompts and suggestions

#	Prompt (summarised)	What the tool suggested
1	What minimum SDK is recommended for a typical Android project, and why?	API 26 (Android 8.0). Reasoning: ~95% device coverage, AndroidX requires 23+ so 23 is a hard floor, and 26 removes dual code paths for notification channels, adaptive icons, and background execution limits.
2	Best way to organise files in the Empty Activity template, and why?	Four source files: <code>MainActivity.kt</code> kept thin, <code>model/MobilityDimension.kt</code> , and <code>ui/</code> holding the screen and components. Advised the data class store <code>@StringRes</code> IDs rather than literal strings so all text stays in <code>strings.xml</code> . Advised against adding a <code>ViewModel</code> or navigation for a single screen.
3	(Attached starter project) What activities and components do I need? Step by step, I am new to Android Studio.	Corrected the plan of using multiple Activities: Compose uses one Activity and many composables, and the assignment requires a single screen. Gave a six-stage build order and a component checklist mapping each assignment requirement to a composable.
4	Should I create a new <code>ui</code> package or reuse <code>ui.theme</code> ?	<code>ui</code> already exists on disk and is hidden by the IDE's "Compact Middle Packages" setting. Advised creating files in <code>ui</code> and keeping <code>ui.theme</code> for theming only.
5	Show how the Stage 1 code is written, so I can see the syntax.	Provided a first-version <code>MobilityLensScreen</code> with a <code>Column</code> and two <code>Text</code> composables, the matching <code>MainActivity</code> , and an annotated explanation of <code>@Composable</code> , trailing lambdas, the <code>modifier</code> parameter convention, <code>stringResource</code> , and <code>dp</code> vs <code>sp</code> .
6	The navigator composable cannot change the index passed into it. How do I fix this?	Explained that Kotlin parameters are always <code>val</code> , and that mutating a copy would not trigger recomposition anyway. Suggested the state hoisting pattern: pass <code>onPreviousClick / onNextClick</code> lambdas and enabled flags instead of the index.
7	I cannot call <code>stringResource</code> inside the button's click handler.	<code>stringResource</code> is a composable and click handlers run after composition. Suggested reading a <code>Context</code> during composition and using <code>context.getString</code> in

		the lambda, with an alternative of capturing pre-read strings.
8	I cannot rename <code>com.example</code> to <code>com.jerinjoseph</code> in the Project panel.	Turn off "Compact Middle Packages" to expose example as its own node, then Refactor → Rename choosing "package". Also update namespace and <code>applicationId</code> in <code>build.gradle.kts</code> manually and rebuild.
9	(Pasted all four composables) Review my UI and make it more readable.	Flagged missing <code>verticalScroll</code> , <code>DimensionCard</code> not using an actual Card, feedback not clearing on navigation, a discarded <code>modifier</code> parameter, and stacked padding. Suggested typography hierarchy, labels above the description and implication, a "Dimension X of 6" indicator, a labelled text field, conditional feedback rendering, and error-coloured feedback. Also advised disabling <code>dynamicColor</code> .
10	I get a warning: "Querying resource values using <code>LocalContext.current</code> ".	Explained it is a lint warning about resources not re-reading on configuration change. Suggested replacing <code>LocalContext.current</code> with <code>LocalResources.current</code> .
11	Best way to set up Git through the Android Studio IDE?	Connect a GitHub account in Settings, then Git → GitHub → Share Project on GitHub, which initialises, commits, creates the remote, and pushes in one step.
12	I want to commit everything except the Gradle builds.	Distinguished build outputs (exclude) from build scripts (must commit). Confirmed the template <code>.gitignore</code> already excludes <code>/build</code> , <code>.gradle</code> , and <code>local.properties</code> , so no change was needed.
13	What do min, target and compile SDK mean, and why do they matter?	<code>compileSdk</code> acts at build time and sets which APIs can be written, <code>minSdk</code> acts at install time and sets the device floor, <code>targetSdk</code> acts at run time and selects which OS behaviours apply.